So, $(x_1, y_1, 0)$ is a point on the required line, which is parallel to $l_1\hat{i} + l_2\hat{j} + l_3\hat{k}$. So, the equation of the line is $\frac{x - x_1}{l_1} = \frac{y - y_1}{l_2} = \frac{z - 0}{l_3}$.

6.8.15 Equation of a plane passing through the line of intersection of two given planes

Theorem 6.22

The vector equation of a plane which passes through the line of intersection of the planes $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ is given by $(\vec{r} \cdot \vec{n}_1 - d_1) + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0$, where $\lambda \in \mathbb{R}$.

Proof

Consider the equation

$$(\vec{r} \cdot \vec{n}_1 - d_1) + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0 \quad \text{... (1)}$$

The above equation can be simplified as

$$\vec{r} \cdot (\vec{n}_1 + \lambda\vec{n}_2) - (d_1 + \lambda d_2) = 0 \quad \text{... (2)}$$

Put $\vec{n} = \vec{n}_1 + \lambda\vec{n}_2$, $d = (d_1 + \lambda d_2)$.

Then the equation (2) becomes

$$\vec{r} \cdot \vec{n} = d \quad \text{... (3)}$$

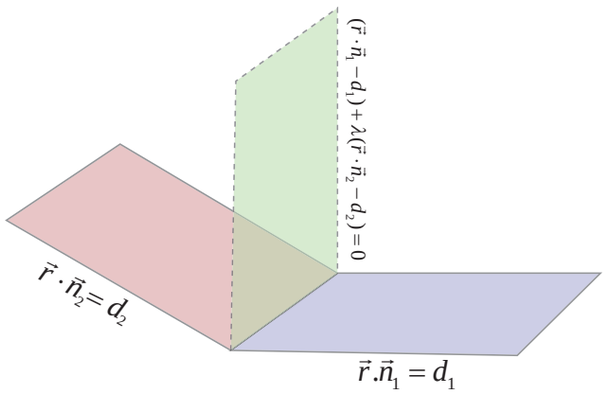


Fig. 6.34

The equation (3) represents a plane. Hence (1) represents a plane.

Let $\vec{r}_1$ be the position vector of any point on the line of intersection of the plane. Then $\vec{r}_1$ satisfies both the equations $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$. So, we have

$$\vec{r}_1 \cdot \vec{n}_1 = d_1 \quad \text{... (4)}$$

$$\text{and} \quad \vec{r}_2 \cdot \vec{n}_2 = d_2 \quad \text{... (5)}$$

By (4) and (5), $\vec{r}_1$ satisfies (1). So, any point on the line of intersection lies on the plane (1). This proves that the plane (1) passes through the line of intersection.

The **cartesian equation** of a plane which passes through the line of intersection of the planes $a_1x + b_1y + c_1z = d_1$ and $a_2x + b_2y + c_2z = d_2$ is given by

$$(a_1x + b_1y + c_1z - d_1) + \lambda(a_2x + b_2y + c_2z - d_2) = 0$$

Example 6.53

Find the equation of the plane passing through the intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) + 1 = 0$ and $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) = 2$ and the point $(-1, 2, 1)$.

Solution

We know that the vector equation of a plane passing through the line of intersection of the planes $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ is given by $(\vec{r} \cdot \vec{n}_1 - d_1) + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0$

Substituting $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $\vec{n}_1 = \hat{i} + \hat{j} + \hat{k}$, $\vec{n}_2 = 2\hat{i} - 3\hat{j} + 5\hat{k}$, $d_1 = 1$, $d_2 = -2$ in the above equation, we get

$$(x + y + z + 1) + \lambda(2x - 3y + 5z - 2) = 0$$

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